Full Harmonic Traces, Quantitative Invariant Graphs, and Slow-Time Growth in the Three-Dimensional Harmonic-Dimension Counterexample

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Abstract

The source manuscript constructs, for each sufficiently large prescribed integer degree, a three-dimensional nonnegative-Ricci metric with more polynomial-growth harmonic functions than Euclidean space. Its mechanism couples a finite angular eigenline cycle to full, positive, whole-ball harmonic restriction operators. We rederive the distinction between full-operator tail separation and higher-accuracy estimates on finitely many columns. We prove quantitative finite-horizon and two-residual invariant-graph bounds, including a conditional nonuniform-product version, and retain the exact infinite trace compatibility needed for entire harmonic functions. The source's original schedule gives an explicit logarithmic growth error of order $T^{3/4}$, rather than only $o(T)$. Conditional on the same finite angular program and its uniform crossing margins, we extend the radial and matching construction to $\eta(t) = t^{-\rho}$, $1/2 < \rho < 1$, with controls of size η^{β}, $1/2 < \beta < 1$. Balancing the resulting proven growth budgets at $\rho = 2/3$ yields an $O(T^{2/3})$ error for the modified metrics. We give every-radius estimates and explain the nonconstructive constants, complementary-mode obligations, and retained global source dependencies. Contribution assessment: C3/5, source-specific conditional graph, matching, and slow-time growth refinements with complete local proofs, not an independent validation of the global counterexample or a historical priority claim.

1 Problem, quantifiers, and source boundaries

For a complete pointed real Riemannian manifold (X, g, o), define

$$\mathcal{H}_k(X) = \{u \in C^{\infty}(X;\mathbb{R}) : \Delta_g u = 0,\ |u(x)| \le C_u(1 + d_g(o,x))^k \text{ for every } x\},$$
$$h_k(X) = \dim_{\mathbb{R}} \mathcal{H}_k(X).$$

On $\mathbb{R}^3$ the degree-l harmonic polynomials have dimension $2l+1$, hence $h_k(\mathbb{R}^3) = (k+1)^2$ for integer $k \ge 0$. The polynomial characterization follows by applying $k+1$ derivatives to a scale-R averaging kernel: its action on a function with degree-k growth is $O(R^{-1})$, so the derivative vanishes as $R \to \infty$. The homogeneous harmonic decomposition then gives the count.

The pinned public source [1] asserts: for every

$$\frac{4}{9} < v < 1, \qquad 1 < c < \frac{9}{4}v, \tag{1}$$

there exists $k_0(v,c)$ such that for each integer $k \geq k_0$ there is a smooth complete metric g_k on $\mathbb{R}^3$, Euclidean near the origin, with $\mathrm{Ric} \geq 0$, asymptotic volume ratio v, and $h_k \geq c(k+1)^2$. Its examples have positive Ricci curvature outside a compact set, nonunique tangent cones, and negative radial sectional planes arbitrarily far out. A further source statement permits arbitrarily small bilipschitz distortion from the Euclidean metric while retaining a ratio $c < 9/4$ by choosing v close enough to one. The metric depends on k. Neither the source statement nor this report interchanges "for every k there is a metric" with "there is one metric for every k."

What is proved here. The angular existence program, elliptic whole-ball Dirichlet theory, and the source's global assertions remain explicitly attributed inputs. Sections 4–6 rederive the analytic mechanisms needed to use those inputs. The quantitative graph theorem and its residual and product variants are independent local results on bounded operators. Section 7 proves a conditional modification of the source construction, not a new angular existence theorem. Section 8 proves the cumulative and every-radius improvements. Reading a proof is not formal verification, and algebraic tests of these additions cannot certify all angular metrics, all controls, or all radii.

Source chain. The source build is the September 26, 2026 manuscript at the locally preserved initial release commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. All ten TeX files in its build were read, including all proof-bearing sections. The correspondence is:

Source file	Mathematical responsibility
`crossings.tex`	Fixed-area gauge; splitting rank; avoidance; retained doubles; parity inversion.
`angular-program.tex`	Finite cyclic permutation and strict averaged frequency margin.
`geometry.tex`	Uniform controlled metric, radial concavity, Ricci Schur estimate, volume and cones.
`transfer.tex`	Full L^2 restrictions, two freezing accuracies, invariant graph, crossing derivative.
`matching.tex`	Stable scalar Green sums, endpoint signs, simultaneous infinite controls.
`growth.tex`	Compatible traces, entire solutions, exponent and pointwise growth.

The master file, introduction, bibliography, and crossing figure were also read. Literal local paths, read intervals, and content hashes are recorded separately in the M693 provenance ledger. No blanket private-corpus read is claimed.

2 The finite angular program and its costs

Let g_0 be the round sphere and $d\mu = d\,\mathrm{vol}_{g_0}/(4\pi)$. For $w \in C^\infty(S^2)$ set $\varrho_w = e^{2w}/\int e^{2w}\,\mathrm{d}\mu$. The conformal metric $\varrho_w g_0$ has area 4π. A smoothly parameterized Moser pullback puts its area form at exactly $4\pi\mu$; this is not replacement by an equivalent measure. Explicitly interpolate $\varrho_{\tau,w} = 1 + \tau(\varrho_w - 1)$, solve $\Delta_0\phi = -(\varrho_w - 1)$ with mean zero, and integrate the vector field $X = \varrho_{\tau,w}^{-1}\nabla_0\phi$. Then

$$\partial_\tau \Phi_\tau^*(\varrho_{\tau,w}\,\mathrm{d}\mu) = \Phi_\tau^*(\partial_\tau \varrho_{\tau,w} + \mathrm{div}_0(\varrho_{\tau,w}X))\,\mathrm{d}\mu = 0.$$

Write $H(w) = \Phi_1^*(\varrho_w g_0)$. On the actual finite parameter families all smooth bounds are uniform; small C^2 size gives small metric distortion by the Poisson estimate and the flow variational equation. The Laplacians then have the common domain $H^2(S^2)$ in $L^2(\mu)$.

2.1 Why individual doubles can be controlled

At an eigenvalue $\lambda > 0$, an orthonormal repeated eigenspace has variation

$$\dot{M}_{ij} = -2\lambda \int u_i u_j \left(\dot{w} - \int \dot{w} \, \mathrm{d}\mu_w \right) \mathrm{d}\mu_w. \tag{2}$$

This follows by differentiating the unchanged two-dimensional Dirichlet form and its changing mass form. The gauge adds a commutator, whose compression to λI is zero. For a pair u, v, the trace-free entries are tested by $u^2 - v^2$ and $2uv$. If these two functions were linearly dependent, (u, v) would lie locally on one of two lines, and a nontrivial linear combination of the eigenfunctions would vanish on an open set. Unique continuation contradicts orthonormality. Their Gram matrix is therefore positive definite. For equal parity these two test directions are even.

The source's finite-cutoff avoidance proof uses compactness of all normalized low-eigenvalue pairs, a finite common set of splitting directions, and a countable family of local codimension-two charts. A chart for an exact multiplicity-m block does not assert that all smaller repetitions in that block satisfy the same two equations. They receive their own charts. Projection of each codimension-two time-parameter chart to parameter space has measure zero; a countable union remains null. This is the needed distinction between full-block multiplicity and one chosen pair.

To retain one double while splitting a different repeated eigenvalue, constrain directions to annihilate its two product functions. If every such direction had scalar compression on the other eigenspace, its pair of product functions would be an invertible constant linear transform of the first pair. Using the squaring map $(x, y) \mapsto (x^2 - y^2, 2xy)$ shows that one eigenfunction from each distinct eigenvalue has the same nodal set. On a connected common nodal domain the two positive restrictions lie in H_0^1: truncations $(u - \epsilon)_+$ give the required approximation without boundary regularity. Testing their equations against each other forces the eigenvalues equal. This contradiction gives a nonscalar constrained direction. A two-variable implicit function argument keeps the chosen double exactly, and the finite integer $\sum \binom{m_j}{2}$ decreases as the remaining clusters split.

2.2 Inside and across round blocks

For round degree l, multiplication is an isomorphism

$$\operatorname{Sym}^2 \mathcal{Y}_l \longrightarrow \bigoplus_{j=0}^{l} \mathcal{Y}_{2j}.$$

Indeed $F_l(z) = (1 - z^2)^l$ is a sum of squares of degree-l harmonics, and

$$(\Delta_0 + 2l(2l+1))F_l = 4l^2 F_{l-1}.$$

The Legendre coefficients at all even degrees are nonzero by induction. Polynomial spectral projections and rotations give surjectivity, and both dimensions equal $(l+1)(2l+1)$. Dualizing (2) gives full trace-free matrix control within the round block.

For an adjacent cross-block pair the source uses the even rotational metric

$$\widehat{H}_l = (1 - c_0\psi/l)^2 g_0, \qquad c_0 = 3/2, \quad 0 \le \psi \le 1, \quad \psi(\pi/2) = 1, \quad \bar{\psi} = \pi^{-1} \int_0^\pi \psi < 1/10.$$

Sector Friedrichs domains are the closures of smooth sphere profiles; at a pole these are $t^{|m|}a(t^2)$ for azimuthal number $m \neq 0$, and even smooth profiles for $m = 0$. The constant Wronskian vanishes

at a pole, proving simplicity in each sector. Positivity of its ground profile fixes reflection parity. The azimuthal $m = l$ ground state concentrates at the equator from $\int(\sin^{-2}\theta - 1)u_l^2 = O(l^{-1})$. Consequently

$$\lambda_{\text{high}} \geq l^2 + (1 + 2c_0)l + o(l).$$

For the axisymmetric slot formerly at degree $l + 1$, the coordinate $x = R^{-1}\int_0^\theta(1 - c_0\psi/l)$ with $R = 1 - c_0\bar{\psi}/l$ and multiplication by $\sqrt{F}$ convert the profile form to the round form plus the bounded potential $((\sqrt{F})'' + \cot x(\sqrt{F})')/\sqrt{F} = O(l^{-1})$. The even endpoint core and its closure are preserved. Thus

$$\lambda_{\text{low}} = l^2 + (3 + 2c_0\bar{\psi})l + O(1), \qquad \lambda_{\text{high}} - \lambda_{\text{low}} \geq (1 - 3\bar{\psi})l + o(l) > 0.$$

Parity counts $(l+1)(l+2)/2$ and $l(l+1)/2$ locate the crossing at combined positions $(l+1)^2, (l+1)^2+1$. Avoidance within each parity and retention of the selected double finish the source construction. These are source arguments, not numerical eigenvalue evidence.

2.3 The frequency margin

Fix $a = \sqrt{v}$, choose $\sqrt{c} < \vartheta < 3a/2$, and a near-round neighborhood with $K_H > \kappa_* > a^2$. Set

$$L = \lfloor\sqrt{k}\rfloor, \quad M = \lfloor\vartheta k\rfloor, \quad p = (M + 1)^2, \quad I = \{(L + 1)^2 + 1, \ldots, p\}.$$

The angular program concatenates the adjacent swaps in I into one cycle, with a positive gap between the whole retained cluster and position $p+1$. Write s for unwrapped phase. A real orthonormal frame $e_i(s)$ can continue through the deliberately constructed simple doubles. The labeled eigenvalues have a finite period P; signs may double the period of the frame. Every pair in the band crosses during the label cycle. Each crossing has a control direction with a nonzero off-diagonal entry. All gaps, slopes, derivatives, and their positive lower bounds here are for a fixed k.

Put $d(x) = (\sqrt{1 + 4x} - 1)/2$ and

$$\Theta_i(s) = d(a^{-2}\lambda_i(s)), \qquad m_i = P^{-1}\int_0^P \Theta_i(s)\,\mathrm{d}s, \qquad \Delta = k - \max_{1\leq i\leq p} m_i > 0. \tag{3}$$

The cyclic permutation makes the band mean the average of the ordered band at every phase, irrespective of unequal durations of the constituent paths. If $\lambda_j(H) \leq C_* l(l + 1)$ on the degree-l positions, then $d(x) \leq \sqrt{x}$ gives

$$m_i \leq \frac{\sqrt{C_*}}{a}\frac{\sum_{l=L+1}^{M}(2l + 1)(l + 1)}{(M + 1)^2 - (L + 1)^2} = \left(\frac{2\vartheta\sqrt{C_*}}{3a} + o(1)\right)k < k.$$

Choose C_* before k so the last coefficient is below one. The uncycled lower modes have $m_i = O(\sqrt{k})$. A single frozen link does not produce this average; continuing labels through the entire cycle does.

3 Ram rederivations on the actual domains

The private sources used here are the original core-pattern reports 5 and 4 and TR207, at the proof locators listed in the bibliography. Their role is finite retained-space reconstruction with explicit complementary error, norm-controlled inversion, and weak operator residuals. Classical Neumann, Schur, frame, and coercive-energy arguments are not renamed as new mathematics.

3.1 Finite measurements and physical norms

Let $\Phi : \mathbb{R}^p \to \mathcal{H}$ be injective, $G = \Phi^*\Phi > 0$, S a bounded measurement map, and $W > 0$ a fixed observation metric. With $A = W^{1/2}S\Phi G^{-1/2}$ and $\sigma_{\min}(A) = s_0 > 0$, the canonical decoder

$$D = G^{-1/2}A^\dagger W^{1/2}$$

satisfies $DS\Phi = \mathrm{I}$ and $\|\Phi Dz\| \le s_0^{-1}\|z\|_W$. If $u = \Phi c + u_\perp$ and observations are $S\Phi c + e$, orthogonality gives the exact identity

$$\|\Phi D(S\Phi c + e) - u\|^2 = \|u_\perp\|^2 + \|\Phi De\|^2 . \tag{4}$$

When the same observations act on the full u, e includes $Su_\perp$, not only noise. This rederives core-pattern 4's restricted dual and core-pattern 5's retained/discarded identity. In the source $\Phi = \iota_j$ is an isometry, but this does not make its finite range all of $L^2(\mu)$.

If $E : \mathbb{R}^p \to \mathcal{H}$ has columns v_i, then

$$\|E\|^2 = \lambda_{\max}\big((\langle v_j, v_i\rangle)_{ij}\big) \le \sum_i \|v_i\|^2 . \tag{5}$$

This is the exact full Gram matrix, not an assumption of independent errors. Finitely many column bounds therefore cost up to $\sqrt{p}$, with p fixed before the late-radius limit. They say nothing about the operator on the orthogonal complement.

3.2 Inversion and two residuals

For $A = D(\mathrm{I} - N)$ with $\|N\| \le q < 1$, multiplication of the finite sum by $\mathrm{I} - N$ leaves $\mathrm{I} - N^{K+1}$. Summing its tail proves

$$\left\| A^{-1} - \sum_{j=0}^{K} N^j D^{-1} \right\| \le \frac{q^{K+1}}{1-q} \left\| D^{-1} \right\| .$$

Factor order and the norm are essential. Spectral radius alone gives no such constant in a prescribed nonnormal norm. For a two-by-two block solve with an invertible eliminated block A_{22}, the reduced operator and source are $S = A_{11} - A_{12}A_{22}^{-1}A_{21}$ and $g = f_1 - A_{12}A_{22}^{-1}f_2$. An approximate pair has

$$r_1 = r_S + A_{12}A_{22}^{-1}r_2.$$

Substituting $\widehat{x}_2 = A_{22}^{-1}(f_2 - A_{21}\widehat{x}_1 - r_2)$ proves it. An accurate reduced row cannot absorb an unbounded complementary residual. Section 5 gives the corresponding exact two-row identity for the moving harmonic graph.

3.3 Weak rows do not close the full trace problem

TR207's integration-by-parts identity is

$$\int \Theta \sum_{n=0}^{N} k_n f^{(n)} = \int f \sum_{n=0}^{N} (-1)^n (k_n\Theta)^{(n)}, \qquad \Theta \in C_c^N .$$

Boundary-reaching atoms add $[\sum_{j=0}^{n-1}(-1)^j f^{(n-1-j)}(k_n\Theta)^{(j)}]$ at the endpoints. Every coefficient and taper derivative is present; symbol freezing alone omits them. A finite coefficient model is identifiable

exactly when its actual response has full column rank. For a square response A replaced by $A+E$, the exact solve obeys

$$\|\widehat{c}-c\| \le \frac{\|e\|+\|E\|\,\|c\|}{\sigma_{\min}(A)-\|E\|}, \qquad \|E\| < \sigma_{\min}(A).$$

This follows from $(A+E)(\widehat{c}-c) = e - Ec$. It is a retained-model bound, not a proof of an infinite-dimensional inverse.

For the actual elliptic problem, core-pattern 4's energy proof uses the full dual norm: if $e \in H_0^1(\Omega)$ and $a(u,v)$ has coercivity $a_0 > 0$, then $a(e,e) = r(e)$ gives $\|\nabla e\|_2 \le a_0^{-1}\,\|r\|_{H^{-1}}$. A boundary lift carries its own error. The scalar coefficient statement extends directly to the uniformly positive definite metric coefficient matrix. Finitely sampled weak rows are only a restricted dual norm. The source instead uses the actual whole-ball Dirichlet operator and full elliptic estimates; there is no sampled-residual shortcut.

4 Full positive harmonic restrictions

Write $r = e^t$ and

$$g = \mathrm{d}r^2 + f(r)^2 H_z(t), \qquad b(t) = e^{-t} f(e^t), \qquad d\,\mathrm{vol}_{H_z} = 4\pi\,\mathrm{d}\mu.$$

Here z is the entire sequence of crossing controls. For $r < R$, let $\mathcal{R}_{r,R}$ mean harmonic extension from ∂B_R into the *whole ball*, restricted to ∂B_r. The regular origin is part of its definition. It is not the propagator obtained by assigning arbitrary data to an inner annular boundary.

Proposition 4.1 (Full trace contraction)**.** *On $\mathcal{H} = L^2(S^2, \mu; \mathbb{R})$, $\mathcal{R}_{r,R}$ extends to a contraction that preserves constants and mean and is order-positive. For positive radial separations its output is smooth in the interior.*

Proof. For smooth boundary data the maximum principle gives order positivity and preservation of 1. The angular average of the harmonic equation satisfies $(f^2 \bar{u}_r)' = 0$. Regularity in the Euclidean core makes the constant radial flux zero; hence the mean is preserved. Positivity and preservation of constants imply $(\mathcal{R}v)^2 \le \mathcal{R}(v^2)$, either by the harmonic measure representation or by applying positivity to $(v-c)^2$ with $c = \mathcal{R}v(x)$. Integrate against μ to obtain the L^2 contraction. Density gives the bounded extension. Interior elliptic estimates on separated cylinders make its images smooth and identify them with the harmonic limits of smooth approximants. □

Here "positive" means order-positive, not a claim that the generally nonstationary restriction is self-adjoint or has a nonnegative quadratic form.

In logarithmic coordinates the equation is

$$u_{tt} + \gamma u_t + b^{-2}\Delta_{H_z(t)} u = 0, \quad \gamma = 1 + 2b_t/b, \qquad \partial_t(e^t b^2 u_t) + e^t \Delta_{H_z(t)} u = 0. \tag{6}$$

Multiplying the divergence form by a constant depending on the cylinder center gives uniform ellipticity on fixed-width cylinders, with $a \le b \le 1$ and uniformly controlled angular geometry. Constants in the resulting estimates depend on the fixed angular program.

4.1 Two freezing statements, two different uses

Let $T_j = \mathcal{R}_{e^j, e^{j+1}}$, $L(s) = -\Delta_{H(s)}$, and

$$P_j^0 = \exp[-d(b(j)^{-2} L(s(j)))], \qquad P_j^* = \exp[-d(b(j)^{-2}(-\Delta_{H_z(j)}))].$$

Suppose $s' = \eta(t) = t^{-\rho}$, $0 < \rho < 1$, and $H_z - H(s) = O(\eta^\beta)$ in the smooth norms needed below, with $H_{z,t} = O(\eta)$, $b \to a$, and $b_t = O(\eta^2)$. The source case is $\rho = \beta = 3/4$; Section 7 establishes the needed geometric hypotheses for the larger stated range.

The full estimate is

$$\left\|T_j - P_j^0\right\|_{\mathcal{H}\to\mathcal{H}} = o(1), \tag{7}$$

uniformly over all controls and allowed earlier histories. Here is the source compactness argument, with its quantifiers retained. If the assertion failed, choose $j_n \to \infty$, admissible histories, and unit inputs $v_n \rightharpoonup v$. Translate the outer radius to $x = 0$. On every fixed left cylinder the coefficients converge smoothly along a subsequence to a frozen angular metric and $b = a$. Inward contraction bounds the solution at every $x < 0$; elliptic compactness makes its restriction at $x = -1$ strongly convergent. The trace at 0 is identified weakly: pairing with each smooth angular test yields a scalar equation $y'' + \gamma y' = f$ with uniformly bounded right side. Boundedness of y gives a bounded derivative at a point of $[-1, -1/2]$, and an integrating factor then gives a uniform derivative bound up to 0. The limit has trace v.

Each limiting angular coefficient solves $y'' + y' - a^{-2}\lambda y = 0$. Boundedness as $x \to -\infty$ excludes the root $-1 - d(a^{-2}\lambda)$, including the root -1 in the constant mode. Thus the limit is $e^{xd(a^{-2}L)}v$. The frozen operators applied to v_n have the same strong interior limit, contradicting failure of (7). Compactness supplies no rate. It does not supply $o(\eta^\beta)$ in the full operator norm.

On the finitely many smooth columns $\phi = e_i(s(j+1))$, a separate barrier gives

$$\left\|(T_j - P_j^*)\phi\right\|_{L^2} \le C\{\eta_j \log(1/\eta_j) + \eta_j^2\} = o(\eta_j^\beta), \qquad \beta < 1. \tag{8}$$

For completeness, use the slab $[j - \ell, j + 1]$ with $\ell = 4\log(1/\eta_j)$. Its length is $o(j)$, so $\eta \asymp \eta_j$. The bounded-backward frozen solution with boundary value ϕ has uniformly bounded smooth norms on this slab, because only a fixed finite smooth family is involved. Inserting it into (6) gives a residual bounded by $C\eta_j(1 + \ell)$, from coefficient variation and the smaller drift error. At the left edge the discrepancy is uniformly bounded by the maximum principle, independently of the earlier metric. For $\gamma \ge 3/4$, a suitable multiple of

$$\eta_j(1 + \ell)(j + 1 - t) + \exp[-(t - j + \ell)/2]$$

is a supersolution for its absolute value. At $t = j$ this gives the displayed bound. Equation (5) converts these finitely many column estimates to a retained-space operator estimate with a fixed-p cost.

Remark 4.2 (Positivity does not repair a missing tail). Let E_n be conditional expectation onto dyadic step functions with 2^n cells in $L^2([0, 1])$. Both E_n and I are order-positive, unital, mean-preserving contractions. Their difference is zero on $\operatorname{ran} E_n$, yet $\|\mathrm{I} - E_n\| = 1$: take a unit function with zero mean on every cell. As $n \to \infty$, $E_n \to \mathrm{I}$ strongly but not in operator norm. Thus even exact finite-frame agreement and strong convergence do not replace the compact smoothing argument for (7).

5 Quantitative graphs with complementary error

Identify the low space at radius j by the isometry $\iota_j c = \sum_i c_i e_i(s(j))$, and put $Q_j = \operatorname{ran}(\iota_j)^\perp$. A general bounded operator from $\mathbb{R}^p \oplus Q_{j+1}$ to $\mathbb{R}^p \oplus Q_j$ has blocks

$$T_j = \begin{pmatrix} L_j & B_j \\ C_j & F_j \end{pmatrix}, \quad L_j = D_j + E_j, \quad \sigma_{\min}(D_j) \ge \tau_j > 0. \tag{9}$$

All norms below are the actual Hilbert norms, not unscaled entrywise maxima. Write e_j, b_j, c_j, f_j for bounds on the norms of E_j, B_j, C_j, F_j, respectively. The symbol b_j here is a block norm, not the radial coefficient $b(t)$.

Theorem 5.1 (Finite-horizon and two-residual graph bounds)**.** *Fix $r_* > 0$. Suppose for every $j \geq J$ that*

$$a_j = \tau_j - e_j - b_j r_* > 0, \qquad \frac{c_j + f_j r_*}{a_j} \leq r_*, \qquad q_j = \frac{f_j + r_* b_j}{a_j} \leq q < 1. \tag{10}$$

Then there is a unique sequence $V_j : \mathbb{R}^p \to Q_j$ in the balls $\|V_j\| \leq r_*$ satisfying

$$V_j = \mathcal{G}_j(V_{j+1}), \qquad \mathcal{G}_j(X) = (C_j + F_j X)(L_j + B_j X)^{-1}. \tag{11}$$

For any terminal graph X_N in its ball, backward iteration to $j < N$ satisfies

$$\|X_j - V_j\| \leq 2r_* \prod_{l=j}^{N-1} q_l. \tag{12}$$

For zero terminal data the constant $2r_*$ can be replaced by r_*. Moreover

$$\|V_j\| \leq \sum_{n=j}^{\infty} \left(\prod_{l=j}^{n-1} q_l \right) \frac{c_n}{\tau_n - e_n}. \tag{13}$$

Empty products equal one. For any candidate graphs X_j in these balls and arbitrary candidate low maps K_j, define the two physical defects

$$R_j^L = L_j + B_j X_{j+1} - K_j, \qquad R_j^H = C_j + F_j X_{j+1} - X_j K_j.$$

For $j < N$ their error is at most

$$\|X_j - V_j\| \leq \sum_{n=j}^{N-1} \left(\prod_{l=j}^{n-1} q_l \right) \frac{\|R_n^H\| + r_* \|R_n^L\|}{a_n} + 2r_* \prod_{l=j}^{N-1} q_l. \tag{14}$$

These estimates require an actual bound on the full complementary block F_j.

Proof. For $\|X\| \leq r_*$, the least singular value of $M_j(X) = L_j + B_j X$ is at least a_j by the triangle inequality. Its inverse therefore has norm at most a_j^{-1}; the second inequality in (10) makes $\mathcal{G}_j$ preserve the ball. Direct multiplication on the right by $M_j(Y)$ yields the noncommutative identity

$$\mathcal{G}_j(X) - \mathcal{G}_j(Y) = (F_j - \mathcal{G}_j(X) B_j)(X - Y) M_j(Y)^{-1}. \tag{15}$$

Indeed $M_j(X) - M_j(Y) = B_j(X - Y)$ and $\mathcal{G}_j(X) M_j(X) = C_j + F_j X$. This proves the Lipschitz constant q_j with no commutation assumption.

Iterate backward from zero at N. Comparing cutoffs $N' > N$ at a fixed j bounds their difference by $r_* \prod_{l=j}^{N-1} q_l \leq r_* q^{N-j}$. Each operator space is complete, so the sequence converges in norm. Continuity gives (11). The same product estimate applied to two infinite solutions and then $N \to \infty$ proves uniqueness and (12). Since $\|\mathcal{G}_j(0)\| \leq c_j/(\tau_j - e_j)$, iterate $\|V_j\| \leq c_j/(\tau_j - e_j) + q_j \|V_{j+1}\|$ to prove (13).

Finally, subtraction in the two rows gives the exact identity

$$X_j - \mathcal{G}_j(X_{j+1}) = (X_j R_j^L - R_j^H) M_j(X_{j+1})^{-1}. \tag{16}$$

Its norm is bounded by the summand in (14). Repeatedly use this defect bound and (15), leaving the terminal graph discrepancy as the last term. This proves the claim. □

Corollary 5.2 (Nonuniform products)**.** *The theorem remains true if the bound $q_j \le q < 1$ is removed, provided the first two inequalities in* (10) *hold and, for each fixed j,*

$$P_{j,N} := \prod_{l=j}^{N-1} q_l \longrightarrow 0.$$

The displayed forcing or infinite residual series is asserted as a finite bound only when it converges. Finite-horizon bounds hold without this additional summability. Parameter continuity of the limit requires the terminal-product convergence to be uniform in the parameters under consideration.

Proof. The Cauchy and uniqueness estimates in the proof use only $r_* P_{j,N} \to 0$, not a common single-step contraction. Iterating the scalar inequalities gives the same sums and products. A finite-head continuous map plus a uniformly small terminal error proves continuity; pointwise terminal decay alone does not prove uniform parameter continuity. □

For example, $q_j = 1 - 1/(j+2)$ gives $P_{j,n} = (j+1)/(n+1)$, with no uniform contraction strictly below one. If the one-step forcing is $O((n+1)^{-1-\epsilon})$, its weighted sum is $O((j+1)^{-\epsilon})$. This illustrates the product criterion only; the separate ball-invariance and inverse-margin hypotheses remain necessary. The source already has a uniform $q < 1$ after fixing k and taking J late, so this corollary is not needed for its proof.

5.1 Specialization to M693 and inverse conditioning

For the source, $D_j = \operatorname{diag}(e^{-d_i(j)})$ with $d_i(j) = d(b(j)^{-2}\lambda_i(s(j)))$ and $\tau_j = \min_i e^{-d_i(j)} \ge \tau_* > 0$. The outer spectral gap and (7) imply, for a fixed $\kappa < 1$,

$$e_j, c_j = O(\delta_j), \qquad b_j = o(1), \qquad f_j \le \kappa\tau_j, \qquad \delta_j = \eta_j^{\beta}. \tag{17}$$

The first two bounds use (8), the $O(\delta_j)$ shift of the whole low-cluster projector, and $O(\eta_j)$ frame movement. Internal doubles cause no loss because the cluster contour is separated from the complement. The other bounds use the full operator estimate. Choose r_* small and then J large to obtain (10). The graph forcing sum gives $V_j = O(\delta_j)$ since δ_j decreases.

The exact low map and its outward inverse are

$$M_j = L_j + B_j V_{j+1}, \qquad A_j = M_j^{-1}.$$

Let $z_j = e_j + b_j \|V_{j+1}\| < \tau_j$. The inverse-difference identity gives

$$\left\|A_j - D_j^{-1}\right\| \le \frac{z_j}{\tau_j(\tau_j - z_j)}, \qquad \left\|M_j^{-1} - L_j^{-1}\right\| \le \frac{b_j \|V_{j+1}\|}{(\tau_j - z_j)(\tau_j - e_j)}. \tag{18}$$

Both follow by writing $X^{-1} - Y^{-1} = -X^{-1}(X - Y)Y^{-1}$. In particular the graph feedback is $o(\delta_j)$, while $A_j = \operatorname{diag}(e^{d_i(j)}) + O(\delta_j)$. No full-operator $o(\delta_j)$ was used. The factors τ_*^{-1} may be very large as k increases. Fixing k before the late start is essential, and no dimension-free numerical conditioning is asserted.

The source already proves strong continuity of each whole-ball restriction in finitely many active controls and norm continuity of its action on the finite moving frame. Backward graph iteration and the uniform terminal bound then give continuity in the product topology on all future controls. This existing continuity is not a new result here. Our additions are the explicit margins, the two-row error budget, and the finite-horizon/product statements.

6 All crossing controls and scalar matching

6.1 The divided difference and its sign

At a crossing let the local perturbation parameter be $\epsilon = z_\nu \eta_\nu^\beta$, with $z_\nu \in [-1, 1]$ and a phase bump equal to one near the double. Write $\zeta = b(j)^{-2}$ to avoid confusing this spectral scaling with the control exponent β. If $q_\zeta(x) = e^{d(\zeta x)}$, then the leading outward entry is

$$(A_j)_{hi} = z_\nu \eta_\nu^\beta m_{hi}(j) + o(\eta_\nu^\beta), \quad m_{hi} = q_\zeta[\lambda_h, \lambda_i]\langle e_h, \dot{L}_\nu e_i\rangle. \tag{19}$$

For distinct arguments $q[x, y] = (q(x) - q(y))/(x - y)$ and at equality $q[x, x] = q'(x)$. In particular, at a double,

$$m_{hi} = e^{d(\zeta\lambda)}\zeta d'(\zeta\lambda)\langle e_h, \dot{L}_\nu e_i\rangle, \qquad d'(x) = (1 + 4x)^{-1/2}. \tag{20}$$

There is exactly one factor ζ. To see the origin of this entry, differentiate the inward multiplier $p_\zeta = e^{-d(\zeta x)}$ on the isolated low cluster and then invert its low matrix. The identity $-q_\zeta(x)q_\zeta(y)p_\zeta[x, y] = q_\zeta[x, y]$ proves (19). Two complementary projector mismatches give an $O(\epsilon^2)$ error because the inward multiplier is a contraction. Frame motion, column freezing, and graph feedback contribute respectively $O(\eta)$, $O(\eta \log(1/\eta))$, and $o(\eta^\beta)$; all are $o(\eta^\beta)$ for $\beta < 1$. This is precisely the finite-cluster derivative, not a derivative of an asserted full-norm small error. Self-adjoint angular variation makes the reversed entry equal. The nonzero chosen angular entry therefore fixes a common sign on a sufficiently small plateau, uniformly over the finite list of crossing types and the compact radial range.

6.2 Exact projective recurrence

Write $A_j = \mathrm{diag}(\alpha_i(j)) + E_j$, $\alpha_i = e^{d_i}$, and normalize each coefficient column as $v_i(j) = e_i + \sum_{h\neq i} u_{hi}(j)e_h$. For a small box $|u_{hi}(j)| \le \eta_j^\sigma$ the denominator $(A_j v_i(j))_i$ is positive. Division gives exactly

$$u_{hi}(j+1) = r_{hi}(j)u_{hi}(j) + F_{hi}(j), \tag{21}$$

$$r_{hi} = \alpha_h/\alpha_i,$$

$$F_{hi} = \frac{E_{hi} + \sum_{l\neq i} E_{hl}u_{li} - r_{hi}u_{hi}(E_{ii} + \sum_{l\neq i} E_{il}u_{li})}{\alpha_i + E_{ii} + \sum_{l\neq i} E_{il}u_{li}}. \tag{22}$$

Thus $F = O(\eta^\beta)$. At an endpoint control on its plateau, $F_{hi} = z_\nu \eta_\nu^\beta m_{hi}/\alpha_i + o(\eta_\nu^\beta)$ uniformly over all coordinates and all other controls. The box terms are $O(\eta^{\beta+\sigma})$; the denominator correction is smaller too. The sign of $\log r_{hi}$ is exactly the sign of $\lambda_h - \lambda_i$ because d is increasing and the radial scale is common.

Lemma 6.1 (Conditional finite-order crossing estimate)**.** *On a bounded phase interval between two consecutive crossing cuts, assume the speed is comparable to η_ν, its total number of steps is $O(\eta_\nu^{-1})$, and a sign-definite scalar gap obeys*

$$|\log r_j| \ge c \min\{|s_j - s_a|^q, |s_b - s_j|^q, 1\},$$

with a matching upper bound near the endpoints. Here $q \ge 1$ is an odd integer for a sign-changing crossing, and all constants are uniform over the declared finite crossing types. A stable product over m consecutive steps is bounded by

$$C\exp(-c\eta_\nu^q m^{q+1}), \qquad \sum_{m\ge 0} C\exp(-c\eta_\nu^q m^{q+1}) \le C'\eta_\nu^{-q/(q+1)}. \tag{23}$$

If the forcing has size $O(\eta_\nu^\beta)$, the stable trial solution has size $O(\eta_\nu^{\beta-q/(q+1)})$. If, on a fixed central plateau, the endpoint-control forcing has fixed nonzero sign and magnitude at least $c\eta_\nu^\beta$, the corresponding two-sided mismatch has that sign and magnitude at least $c'\eta_\nu^{\beta-q/(q+1)}$ for small enough η_ν.

Proof. For any m consecutive grid points, apart from a bounded number caused by rounding the cuts, at least a fixed fraction are at phase distance at least $c\eta_\nu m$ from the nearer interval endpoint. Sum the lower gap bound over those points to get $c\eta_\nu^q m^{q+1}$; absorb the bounded omissions into C. The stable direction uses r when $r<1$ and r^{-1} when $r>1$, so the products are positive and decay. Comparing the sum with the integral of $e^{-c\eta_\nu^q x^{q+1}}$ and rescaling $x=\eta_\nu^{-q/(q+1)}y$ proves the second assertion. Multiply by the forcing bound.

Near the crossing, the upper gap bound makes the weights on $|j-t_\nu|\le\theta\eta_\nu^{-q/(q+1)}$ bounded below by a fixed positive constant. This window lies inside the fixed phase plateau for late crossings. It contains a fixed multiple of $\eta_\nu^{-q/(q+1)}$ steps. Their forcing contributions have the same sign. The rest of the plateau has that sign as well. Outside it, a product has crossed a fixed phase distance; summing the gap there produces c/η_ν, hence the absolute outside contribution is at most $C\eta_\nu^{\beta-1}e^{-c/\eta_\nu}$, negligible compared with the central contribution. This proves the mismatch statement. □

For the source's simple doubles $q=1$. The local extension to $q>1$ is conditional on the stated gap, phase, and nonzero-control assumptions; no higher-order angular program is constructed here. Choose

$$\frac{1}{2}<\beta<1,\qquad 0<\sigma<\beta-\frac{1}{2} \tag{24}$$

for $q=1$. The trial bound $C\eta^{\beta-1/2}$ then fits strictly inside the box η^σ. A conditional order-q version needs instead $q/(q+1)<\beta<1$ and $0<\sigma<\beta-q/(q+1)$.

6.3 Why one scalar condition suffices at each double

Set the cut $N_\nu=\lceil t_\nu\rceil$, preserving the side of the crossing of each multiplier, with a possible neutral step if it occurs at an integer. For a $+\to-$ sign change of $\log r$, put the coordinate zero at the cut and solve backward on the left and forward on the right. Both directions are stable. For a $-\to+$ change, solve forward from the previous zero cut a and backward from the next zero cut b. The only obstruction is

$$D_\nu=\sum_{q=a}^{N_\nu-1}F_q\prod_{l=q+1}^{N_\nu-1}r_l+\sum_{q=N_\nu}^{b-1}F_q\prod_{l=N_\nu}^{q}r_l^{-1}. \tag{25}$$

The reverse ordered pair has the opposite stability pattern. Consequently a double creates one, not two independent, scalar conditions. All weights in (25) are positive. Orient each control so that $D_\nu>0$ at $z_\nu=1$ and $D_\nu<0$ at $z_\nu=-1$, using Lemma 6.1. Noncrossing pairs have uniform geometric stable sums. For a forward sum, the polynomial ratio of earlier and current η^β is absorbed by the geometric weight; backward sums are easier because η decreases.

For the entire countable collection, define a map on the product of coordinate and control boxes: each new coordinate is its stable trial solution and each new control is $\mathrm{proj}_{[-1,1]}(z_\nu-D_\nu)$. Each coordinate is continuous in the product topology. Finite sums are continuous; the remaining scalar tails have uniform stable bounds, and the graph dependence on the whole control sequence has the terminal estimate of Section 5. On a finite set of coordinates hold the rest at the centers and apply Brouwer to the resulting finite box. A diagonal subsequence in the countable compact product

converges. Continuity makes every fixed coordinate satisfy its equation in the limit. Endpoint signs exclude an endpoint fixed control; for an interior fixed point the projection formula forces $D_\nu = 0$. Thus all scalar recurrences hold simultaneously for one common control sequence, not one metric per eigenline.

At that solution,

$$A_j v_i(j) = \sigma_i(j) v_i(j+1), \qquad \sigma_i(j) = (A_j v_i(j))_i > 0, \quad \log \sigma_i(j) = d_i(j) + O(\eta_j^\beta). \tag{26}$$

The matrix of columns is $\mathrm{I} + O(p\eta_j^\sigma)$ and is invertible for fixed p and sufficiently late J. None of these arguments requires $\sum_j \eta_j^\beta < \infty$.

7 A conditional slow-time extension

Theorem 7.1 (Radial and matching power family)**.** *Fix one source angular program from Section 2, with its smooth finite family, strict outer gap, simple isolated doubles, nonzero control entries, and strict means $m_i < k$. For any*

$$\frac{1}{2} < \rho < 1, \qquad \frac{1}{2} < \beta < 1, \tag{27}$$

replace the source schedule by

$$\eta(t) = t^{-\rho}, \qquad s(t) = \frac{t^{1-\rho} - J^{1-\rho}}{1-\rho}, \qquad \delta_\nu = \eta(t_\nu)^\beta.$$

There is a sufficiently late J, depending on this fixed angular program and these exponents, for which the source radial construction and simultaneous matching work with these powers. In particular they give the same number p of exactly compatible entire harmonic functions on the modified metric. The metric has $\mathrm{Ric} \geq 0$*, the same volume ratio a^2, and the same link family of tangent cones. No uniformity in k, no computed value of J, and no independently constructed angular program are asserted.*

Proof. We verify the geometric, transfer, and matching requirements rather than changing the exponent only in the final formula. Extend the reference angular path smoothly to round for $s \leq -2$, stationary near its start. Use the same disjoint fixed-width phase control bumps and directions as the source:

$$H_z(t) = H\left(w(s(t)) + \sum_\nu z_\nu \eta(t_\nu)^\beta \chi_\nu(s(t))\psi_\nu\right).$$

There is no finite accumulation of supports. On every fixed phase interval $t - t_\nu = O(t_\nu^\rho) = o(t_\nu)$, so $\eta(t)/\eta(t_\nu) \to 1$ uniformly. Since $|\eta'| = \rho t^{-\rho-1} = O(\eta^2)$ for $\rho < 1$, the finite family and its bumps give

$$H_z - H(s) = O(\eta^\beta), \qquad H_{z,t} = O(\eta), \qquad H_{z,tt} = O(\eta^2), \qquad \mathrm{tr}(H_z^{-1} H_{z,t}) = 0. \tag{28}$$

These statements hold with the fixed spatial derivatives required by the elliptic estimates. All higher fixed time derivatives also tend to zero on translated bounded cylinders. Taking J large puts every perturbed parameter in the same angular neighborhood and retains $K_{H_z} > \kappa_* > a^2$.

Choose a nonnegative smooth $\omega(r)$ supported in $(1,2)$ with integral one. Let $\theta(t/J)$ be zero for $t \leq J/2$ and one for $t \geq 7J/10$. For $C > 0$ chosen below define

$$f''(r) = -m_J \omega(r) - C r^{-1} \theta((\log r)/J)(\log r)^{-2\rho}, \tag{29}$$
$$m_J = 1 - a - C\int_{J/2}^\infty \theta(t/J) t^{-2\rho}\,\mathrm{d}t, \quad f(0) = 0, \quad f'(0) = 1.$$

The tail term is defined as zero before its support, so no logarithm at $r = 0$ is involved. The integral is finite precisely because $2\rho > 1$ and is $O(J^{1-2\rho})$. Thus $m_J > 0$ for late J, $a \le f' \le 1$, and $f' \to a$. For $t \ge 7J/10$,

$$b_{tt} + b_t = -Ct^{-2\rho} = -C\eta^2, \qquad b + b_t = f'(e^t) = a + \frac{C}{2\rho - 1} t^{1-2\rho}. \tag{30}$$

The integrating-factor solution for b_t consists of an exponentially decaying initial term and the convolution

$$-C \int_{7J/10}^{t} e^{-(t-v)} v^{-2\rho} \, \mathrm{d}v.$$

For $v \ge 7J/10$, the inequality $\log(t/v) \le (t - v)/(7J/10)$ bounds this convolution by a constant times $t^{-2\rho}$ once J is large. After $t \ge 4J/5$ the initial term is also bounded by this quantity. Hence

$$|b_t| + |b_{tt}| \le C_\rho C\eta^2, \qquad 0 \le b - a \le C_\rho C t^{1-2\rho}, \qquad a \le b \le 1. \tag{31}$$

The first angular motion at $s = -2$ occurs at $t = J - 2J^\rho + o(J^\rho) > 4J/5$, so it is wholly inside this tail region.

For clarity, the actual Ricci calculation is retained. Set $\mathcal{A} = H_z^{-1} H_{z,t}$, $\mathcal{B} = \mathcal{A}/2$, and $q_0 = 1 + b_t/b$. The shape operator is $r^{-1}(q_0 \mathrm{I} + \mathcal{B})$. In the radial and angular orthonormal splitting,

$$\begin{aligned} r^2 \operatorname{Ric}_{rr} &= -2(b_{tt} + b_t)/b - \operatorname{tr}(\mathcal{A}^2)/4, \\ r^2 \operatorname{Ric}_{rT} &= b^{-1} \operatorname{div}_{H_z} \mathcal{B}, \\ r^2 \operatorname{Ric}_{TT} &= (K_{H_z}/b^2 - 2q_0^2 + q_0 - \partial_t q_0)\mathrm{I} + (1 - 2q_0)\mathcal{B} - \partial_t \mathcal{B}. \end{aligned} \tag{32}$$

The tangential block is at least $\tau \mathrm{I}$ for a fixed $\tau > 0$ after increasing J. The mixed block is bounded by $M_M \eta$, and $\|\mathcal{A}\|_{\mathrm{HS}} \le M_A \eta$, with angular constants independent of the controls. Choose C before J so that

$$2C > M_A^2/4 + M_M^2/\tau.$$

As $b \le 1$, the radial Schur complement is then strictly positive in the moving region. Before angular motion the round concave warped metric has nonnegative Ricci curvature. This proves the curvature assertion without assuming that radial sectional curvature is nonnegative.

The radial distance is exactly r: any path has length at least its total radial displacement and the radial path realizes it. Compact coordinate balls and the smooth Euclidean core give completeness. Since the angular area is exactly 4π, $\operatorname{vol} B_R = 4\pi \int_0^R f(r)^2 \, \mathrm{d}r \sim 4\pi a^2 R^3/3$. For any phase s_* choose $t_n \to \infty$ with that phase modulo the period. On each fixed rescaled annulus $s(t_n + \log u) - s(t_n) \to 0$, controls tend to zero, and $b \to a$. The smooth annular limit is $\mathrm{d}u^2 + a^2 u^2 H(s_*)$. The region $u \le \epsilon$ has diameter at most 2ϵ via the origin, so these annular limits also give pointed cone limits. Different link spectra give the same uncountable family of nonisometric cones as in the source. The possible small metric distortion is retained by the same uniform angular neighborhood and a.

The full freezing proof of Section 4 needs only translated smooth convergence and full contraction, both now established. Its column proof uses $\ell = 4 \log(1/\eta)$, and still yields $O(\eta \log(1/\eta)) = o(\eta^\beta)$ for $\beta < 1$. The fixed spectral gap therefore gives (17) and the invariant graph. The crossing derivative (19) retains its nonzero sign and uniform remainder. The simple gap has the same phase slope and the speed is comparable on each interval, so Lemma 6.1 with $q = 1$ gives size $O(\eta^{\beta - 1/2})$. Choose σ as in (24); the simultaneous fixed-point construction and exact cocycle (26) follow. The compatible extension to entire functions is proved in Section 8, completing the asserted analytic construction. □

The restrictions on the exponents have separate meanings. At $\rho = 1/2$ the chosen Ricci-buffer tail has divergent total slope loss. At $\rho = 1$ a bounded phase interval need not have $o(t)$ length and the phase averaging estimate below no longer has a sublinear power. At $\beta = 1/2$ the Green amplification no longer gives a vanishing matching coordinate, whereas $\beta = 1$ loses the strict column-freezing separation. These are limits of this proof, not impossibility theorems for other constructions.

8 Infinite compatibility and every-radius growth

8.1 Exact entire functions, not finite compressions

Let

$$w_i(j) = \iota_j v_i(j) + V_j v_i(j), \qquad c_i(J) = 1, \quad c_i(j) = \prod_{l=J}^{j-1} \sigma_i(l), \qquad g_i(j) = c_i(j) w_i(j).$$

The graph relation and (26) give exactly

$$T_j w_i(j+1) = \sigma_i(j)^{-1} w_i(j), \qquad T_j g_i(j+1) = g_i(j). \tag{33}$$

Thus even an initially L^2 graph trace at radius j is smooth, by representing it as the inward restriction from radius $j+1$. Solve the whole-ball Dirichlet problem with boundary value $g_i(j)$. Uniqueness and (33) make these solutions agree on all smaller balls. Their union is a smooth entire harmonic function U_i. The p functions are independent because their low projections at J are the invertible coefficient columns $v_i(J)$. A family of finite approximate traces with small defects would not give this identity or independence on the infinite union without additional estimates and a limit argument.

Since V_j is orthogonal to the low space and $v_i(j)$ has its ith coordinate equal to one, the norms of $w_i(j)$ are bounded above and below by positive constants. Hence

$$\log \|g_i(n)\| = \sum_{j=J}^{n-1} d_i(j) + O\left(1 + \sum_{j=J}^{n-1} \eta_j^{\beta}\right). \tag{34}$$

The scalar errors may have either sign. Their cumulative bound, not absolute summability over the infinite sequence, is what controls growth.

Theorem 8.1 (Explicit logarithmic growth budget)**.** *For the fixed angular program and modified metrics in Theorem 7.1, there are constants A_0, A_1, A_2, A_3, A_4 such that, for every real $T \geq J + 2$,*

$$\left| \log \left\| U_i(e^T, \cdot) \right\|_{L^2(\mu)} - m_i T \right| \leq A_0 + A_1 T^{\rho} + A_2 T^{2-2\rho} + A_3 T^{1-\rho\beta} + A_4 T^{1-\rho}. \tag{35}$$

The constants can be chosen jointly over the finite list of i and the admitted control sequence. With

$$\chi = \max\{\rho, 2 - 2\rho, 1 - \rho\beta, 1 - \rho\} < 1,$$

one also has the pointwise bound, for all sufficiently large r and every angle,

$$|U_i(r, \omega)| \leq C_i r^{m_i} \exp(C_i (\log r)^{\chi}). \tag{36}$$

In particular $U_i \in \mathcal{H}_k$ for every i. For the source's original $\rho = \beta = 3/4$, the four powers in (35) are $3/4, 1/2, 7/16, 1/4$. For the modified choice $\rho = 2/3$ and any $1/2 < \beta < 1$, one has $\chi = 2/3$.

Proof. The zero-mean periodic function $\Theta_i - m_i$ has a bounded periodic primitive Q_i. Since $s' = t^{-\rho}$, integration by parts gives

$$\int_J^T (\Theta_i(s(t)) - m_i)\,\mathrm{d}t = [t^\rho Q_i(s(t))]_J^T - \rho \int_J^T t^{\rho-1} Q_i(s(t))\,\mathrm{d}t = O(T^\rho) + O_J(1). \tag{37}$$

The phase derivative of Θ_i is bounded for this fixed labeled program, including the smoothly continued double crossings. On $[j, j+1]$ its variation is $O(\eta_j)$, so replacing this integral by the unit-step sum costs $O(T^{1-\rho})$.

For $x \geq 0$, $d'(x) \leq 1$. The finitely many angular eigenvalues are bounded, and $a \leq b \leq 1$ implies $|b^{-2} - a^{-2}| \leq 2a^{-3}|b - a|$. Consequently (31) gives

$$|d(b(j)^{-2}\lambda_i) - d(a^{-2}\lambda_i)| \leq C|b(j) - a| \leq Cj^{1-2\rho}.$$

Its cumulative sum is $O(T^{2-2\rho})$. This explicit estimate is stronger than merely applying Cesaro convergence to $b \to a$. The scalar error in (34) sums to $O(T^{1-\rho\beta})$ because $0 < \rho\beta < 1$. Add the four budgets to prove (35) at integer T.

If $j \leq T \leq j+1$, whole-ball contraction in both comparisons gives

$$\|g_i(j)\| \leq \left\|U_i(e^T, \cdot)\right\| \leq \|g_i(j+1)\|.$$

The integer bound on both sides and the bounded shift of $m_i T$ prove the statement for *every* real T. Thus the spherical logarithmic exponent equals m_i along the full radius variable, not only a selected sequence of radii or phases.

On a fixed-width logarithmic cylinder, uniformly elliptic interior L^2-to-sup estimates for (6) bound the pointwise value by the neighboring cylinder's L^2 norm. Integrating the just-proved spherical upper bound over that fixed width changes only constants, since m_i is fixed and $T \mapsto T^\chi$ changes by a bounded amount there. This proves (36). As $\chi < 1$ and $m_i \leq k - \Delta$, the right side is eventually bounded by $Cr^{k-\Delta/2}$, and a compact-region maximum supplies the remaining radii in the definition of $\mathcal{H}_k$.

For $\rho = 2/3$, both the averaging and radial exponents equal $2/3$. Since $\beta > 1/2$, $1 - 2\beta/3 < 2/3$, and $1 - \rho = 1/3$. Thus $\chi = 2/3$. Among the exponents in this particular budget, $\max(\rho, 2 - 2\rho)$ is minimized at $2/3$; no optimality among other constructions or estimates is claimed. □

Corollary 8.2 (A conditional radius threshold)**.** *Suppose the constants in* (35) *are known, write its four nonconstant powers as* $\chi_l < 1$ *with coefficients* A_l*, and let* Δ *be as in* (3)*. Then for*

$$T \geq T_* := \max\left\{J + 2, 1, \max_{1 \leq l \leq 4}\left(\frac{8A_l}{\Delta}\right)^{1/(1-\chi_l)}\right\} \tag{38}$$

one has $\log\left\|U_i(e^T)\right\| \leq A_0 + (k - \Delta/2)T$*. The physical threshold is* $r_* = \exp T_*$*, with an additional fixed cylinder width when using the pointwise bound.*

Proof. Each nonconstant term is at most $\Delta T/8$, so their sum is at most $\Delta T/2$. Insert $m_i \leq k - \Delta$. This explains the factor eight: four separately allocated errors are being charged against half the gap. □

This is an explicit formula *conditional on actual constants.* The source's nonquantitative full-operator compactness threshold and angular avoidance choices have not been numerically enclosed, so this report does not output a certified numerical value of r_*. Substituting sampled estimates for these constants would change the claim.

Proposition 8.3 (Cumulative scalar-error criterion)**.** *More generally, assume exact trace compatibility, upper and lower bounds on the normalized graph columns, and* $\log \sigma_i(j) = d_i(j) + \varepsilon_i(j)$. *If*

$$\sum_{j=J}^{n-1} d_i(j) = m_i n + o(n), \qquad \left| \sum_{j=J}^{n-1} \varepsilon_i(j) \right| = o(n),$$

then the spherical exponent is m_i *for every radius; the uniform cylinder estimate gives membership in* $\mathcal{H}_k$ *whenever* $m_i < k$. *Absolute summability of* ε_i *is unnecessary.*

Proof. Take the logarithm of the exact product of the positive σ_i, add the bounded normalized-column term, and use the two cumulative assumptions. Bracket intermediate radii by contraction as above, then use the fixed-width elliptic estimate and the strict gap $k - m_i$. □

This proposition weakens only the scalar growth accounting after the graph and all matching controls exist. It does not permit arbitrary large errors in the off-diagonal matching argument or remove the pointwise $o(\delta)$ sign requirement there.

9 Examples, advantages, and limits

9.1 A graph budget with both rows visible

Consider bounds $\tau = 0.7$, $e = 0.01$, $b = 0.02$, $c = 0.005$, $f = 0.2$, $r_* = 0.05$. Then $a = 0.689$, $(c + fr_*)/a < 0.022 < r_*$, and $q = (f + r_* b)/a < 0.292$. These numbers are an illustrative bounded-operator fixture, not estimates for an angular program. A low-row error 10^{-4} and complementary-row error 10^{-5} give a graph defect at most $(10^{-5} + 0.05 \cdot 10^{-4})/0.689$. Even exact low rows leave the complementary term. Conversely, with exact two-row iteration and zero terminal graph, a horizon $N - j$ bounds the graph error by $r_* q^{N-j}$. A measured finite matrix without a proven f bound cannot instantiate the theorem for the continuum operator.

9.2 Original and modified schedules

The source schedule has $\eta = t^{-3/4}$, controls $\delta = t^{-9/16}$, crossing Green length $t^{3/8}$, and trial coordinates of size $t^{-3/16}$. Its box $\eta^{1/8} = t^{-3/32}$ is larger for late t, as required. The scalar errors are not summable: $\sum_{j \le n} j^{-9/16} = O(n^{7/16})$, but this is sublinear.

For the modified choice $\rho = 2/3$, $\beta = 3/4$, choose $\sigma = 1/8$. Then $\delta = t^{-1/2}$, the Green length is $t^{1/3}$, the trial size is $t^{-1/6}$, and the box is $t^{-1/12}$. The four growth budgets are respectively $T^{2/3}, T^{2/3}, T^{1/2}, T^{1/3}$. The improved subpower factor comes with a *slower* convergence of b to a, namely $O(t^{-1/3})$ instead of $O(t^{-1/2})$, balanced against faster phase averaging. It is a new schedule for a conditionally available angular program, not a better estimate on the unchanged source metric.

9.3 What the rederivation adds

The quantitative graph formulas make the roles of full tail separation, column accuracy, inverse conditioning, and terminal truncation distinct. They give a meaningful tolerance allocation if continuum block bounds are eventually supplied. The two-residual formula prevents an apparently accurate low recurrence from concealing an inaccurate complementary graph. The nonuniform-product variant identifies exactly which infinite-horizon conclusion survives without a common contraction factor. The radial family and cumulative calculation replace an unquantified final $o(T)$ by source-specific sublinear powers and preserve every-radius, all-column, one-metric compatibility.

There are costs. The finite low dimension is about k^2; the number of crossing types, smooth norms, cluster gaps, and the inverse margin may deteriorate badly with k. The entire construction uses infinitely many controls and exact whole-ball operators. Finite-horizon computation can approximate a graph only after a full tail margin is established; it cannot itself authenticate the infinite geometry. The optimized exponent $2/3$ is merely the balance of the present radial and phase estimates. The positive Ricci Schur margin is not a sectional-curvature margin. A metric close to Euclidean in bilipschitz norm is not thereby locally conformally flat.

9.4 Future extensions and specific obstacles

A genuinely numerical realization would first need enclosures for the actual angular spectral projectors, outer gaps, splitting entries, and coefficient derivative bounds, followed by an effective full-operator freezing estimate. The compactness proof alone cannot supply those constants. The graph residual formula then allows finite-prefix tolerances with an explicitly priced complementary tail.

Higher odd-order crossings would use Lemma 6.1 and the stricter $\beta > q/(q+1)$ threshold, but an angular path with exactly those degeneracies, nonzero controls, isolation, and uniform phase gaps must be constructed separately. Nonuniform graph products can accommodate deteriorating separation only when their products decay and the weighted defect series converges; positivity or invertibility of isolated low matrices does not ensure either property. Finally, a single metric exhibiting a family of large-degree violations would require compatible angular programs and quantitative estimates across changing k. Nothing in the fixed-k analysis provides them.

10 Current public context and verification scope

The public status check on October 8, 2026 inspected the OpenAI repository, its family-361 scope note, and primary arXiv pages. The repository describes results at different verification stages and currently lists 719 manuscripts. The local manuscript number M693 is a preserved initial-release identifier, not a claim that the current catalogue still has 722 entries. A direct public API request returned current commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`; this status observation does not by itself authenticate each local source byte.

The family-361 public formalization note [5] describes the higher-dimensional companion: dimension 16, degree 50000, and no separate tangent-cone or cone-spectrum conclusions. Its comparator is `HarmonicGrowth.lean`. That is positive public evidence of a related formalization, not a statement that formalization is absent. The complete public comparator text was also inspected at the observed commit: its metric coefficients are smooth and positive definite, its distance is intrinsic, and its harmonic-growth condition is a pointwise bound at every point. Its `MainClaim` quantifies an even dimension at least eight and an integer degree at least two. The comparator declares `axiom mainStatement : MainClaim` and wraps it in a theorem; that interface file is not itself the implementing proof. The scope note does not establish the three-dimensional source's theorem. No Lean kernel run was performed for this report, and the family-level assertion is not promoted to an independently kernel-checked M693 result.

Cai–Lai's August 2026 preprint [6], Theorem A, assumes a complete noncompact *locally conformally flat* manifold of dimension at least three with nonnegative Ricci curvature. It states the Euclidean dimension upper bound for every $d \geq 0$, with Euclidean rigidity at positive integer equality. Its statement was checked in the primary full-text introduction; its proof was not independently audited here. Local conformal flatness is an additional geometric hypothesis, not a consequence of small bilipschitz distortion. Thus this comparison does not directly apply to a general metric in M693.

Lin–Wang–Xu [7], Theorem 1.12, state the integer-degree three-dimensional comparison under *nonnegative sectional curvature* and positive asymptotic volume ratio. Their Theorem 1.8 concerns the full positive spectrum of a smooth two-sphere with $K \geq 1$. These primary theorem statements were checked, not their full proofs. Huang [8] assumes a unique tangent cone together with nonnegative Ricci curvature and maximal volume growth for the cone-spectrum upper bound. Its primary abstract was checked. M693's negative radial sectional planes and nonunique cones distinguish precisely these hypotheses. A single frozen-link counting bound cannot substitute for the changing-label construction.

The local additions use standard spectral calculus, graph transforms, Green sums, and energy estimates. No historical priority search can be inferred from checking these nearby primary papers, and no priority claim is made. The source's own bibliography supplies classical background, including Kato's perturbation theory, Moser's volume-form gauge, and elliptic estimates; this report does not claim a full new audit of those books or of every cited paper.

Evidence limits. The accompanying M693 evidence binds the TeX, PDF, executed bounded tests, literal source/private read scopes, public status record, and visual/build checks to their hashes. Exact finite algebra checks and floating-point bounded examples have different statuses and are reported separately. They do not prove the continuum gap bounds or the global counterexample. The independent parent review, if supplied, is separate from this author's read ledger. A clean PDF is a layout result, not a mathematical validation.

References

[1] OpenAI, *A Three-Dimensional Counterexample to Integer-Degree Harmonic Dimension Comparison*, September 26, 2026. Pinned source, master and all mathematical sections: public manuscript directory.

[2] M. Subbarao, private original `TR-core-pat-5.tex`, core pattern 5. Literal passages read: lines 145–441, including `thm:neumann`, `thm:frame`, `prop:discarded`, `prop:schurresidual`, and `thm:woodbury`. Used for conditional finite inverses, retained-space reconstruction, and complementary residual accounting. Confidential local source.

[3] M. Subbarao, private original `TR-core-pat-4.tex`, core pattern 4. Literal passages read: lines 132–163, 253–430, and 601–682, including `thm:dual`, `prop:panel`, and `thm:energy`. Used with physical metrics, aliasing, full dual norms, and boundary lifts. Confidential local source.

[4] M. Subbarao, private original *TR207: robust calibration and inversion*, `TR207-rmt-robust-calibration-inversion.tex`. Literal passages read: lines 161–333 and 370–505, including `ass:regularity`, `thm:weak`, `prop:remainder`, `thm:rank`, and `thm:perturb`. Confidential local source.

[5] OpenAI, *Failure of integer-degree harmonic dimension comparison*, public family-361 formalization scope note, inspected October 8, 2026. `https://github.com/openai/math/blob/main/lean/docs/361.md`.

[6] X. Cai and M. Lai, *Dimension of polynomial growth harmonic functions on locally conformally flat manifolds with nonnegative Ricci curvature*, arXiv:2608.04553v2 (August 16, 2026), Theorem A. `https://arxiv.org/html/2608.04553v2`.

[7] S. Lin, H. Wang, and G. Xu, *Eigenvalues on spheres*, arXiv:2607.11544v1 (2026), Theorems 1.8 and 1.12. https://arxiv.org/html/2607.11544v1.

[8] X.-T. Huang, *Harmonic functions with polynomial growth on manifolds with nonnegative Ricci curvature*, Calc. Var. PDE **62** (2023), 111; primary preprint arXiv:2109.07534. https://arxiv.org/abs/2109.07534.